

SLIDE 1: SMART INDIA HACKATHON 2026 TITLE

- **Problem Statement ID:** SIH-2026-DISASTER-01
- **Problem Statement Title:** Hyper-Local Flash Flood & Landslide Early Warning System
- **Theme:** Disaster Management / ClimateTech / GeoTech / AI & IoT
- **PS Category:** Software + Hardware / IoT Mesh Integration
- **Team Name:** Team Revoira
- **Solution:** BhoomiRakshak Early Warning Infrastructure

SLIDE 2: IDEA, PROBLEM SOLUTION AND UVP

- **IDEA / SOLUTION:** BhoomiRakshak is an AI & IoT-Powered Hyper-Local Flash Flood & Landslide Early Warning Infrastructure for High-Altitude Himalayan River Basins.
- **Problem Resolution:** Provides 30-120+ minutes of evacuation lead time, automates 110dB sirens and ward SMS broadcasts, and directs populations to high-altitude shelters.
- **Unique Value Propositions (UVP):** Ward-level precision, Dual-hazard risk scoring (Flash Flood + Landslide), and 100vh Single-Window GIS Mission Control UI.

SLIDE 3: TECHNICAL APPROACH & METHODOLOGY

- **1. Telemetry Ingestion:** 24GHz Ultrasonic Stage Gauges, OTT ARG Rain Gauges, TDR Soil Saturation Probes.
- **2. Risk Engine & Fusion:** Real-time Dual-Hazard Scoring (Flash Flood Inundation Index + Slope Shear Index).
- **3. Actionable Evacuation Lead Time:** Dynamic countdown estimation to river crest/breach.
- **4. Automated Warning Broadcast:** 110dB Siren Trigger + Ward SMS Broadcast + IVR Calls + SDRF Dispatch.
- **5. Spatial GIS Dashboard:** Leaflet GIS map with 30-Day and 1s Live Telemetry trend graphs.

SLIDE 4: IMPACT AND BENEFITS

- **Impacts:** Saves lives in vulnerable mountain wards, protects dams and roads, and standardizes national response protocols.

- **Benefits:** Actionable countdown lead time (e.g. '38 Mins Remaining'), multi-channel emergency broadcast, and proactive stress-test simulation.

SLIDE 5: FEASIBILITY AND VIABILITY

- **Feasibility:** Built on Leaflet GIS, Chart.js, React, and standard CWC/IMD/ISRO API telemetry integrations.
- **Viability:** Solar-powered LoRaWAN mesh nodes ensure low maintenance; scalable nationwide across mountain and coastal regions.
- **Risks & Mitigation:** Hardened IP68 radar sensors mounted on elevated bridges prevent flood damage; dual LoRaWAN + Satellite uplinks maintain connectivity.

SLIDE 6: RESEARCH AND REFERENCES

- **Guidelines:** NDMA Manual for Management of Floods & Urban Floods; CWC Automated Observation Station Manual.
- **Research Papers:** Himalayan Flash Flood & GLOF Analysis (ScienceDirect); IoT Landslide Early Warning Systems (IEEE).
- **Official Portals:** Central Water Commission Telemetry (cwc.gov.in); IMD AWS Portal (mausam.imd.gov.in); ISRO MOSDAC (mosdac.gov.in).